Cryptography

Public-Key Cryptography

Cristhian Iza

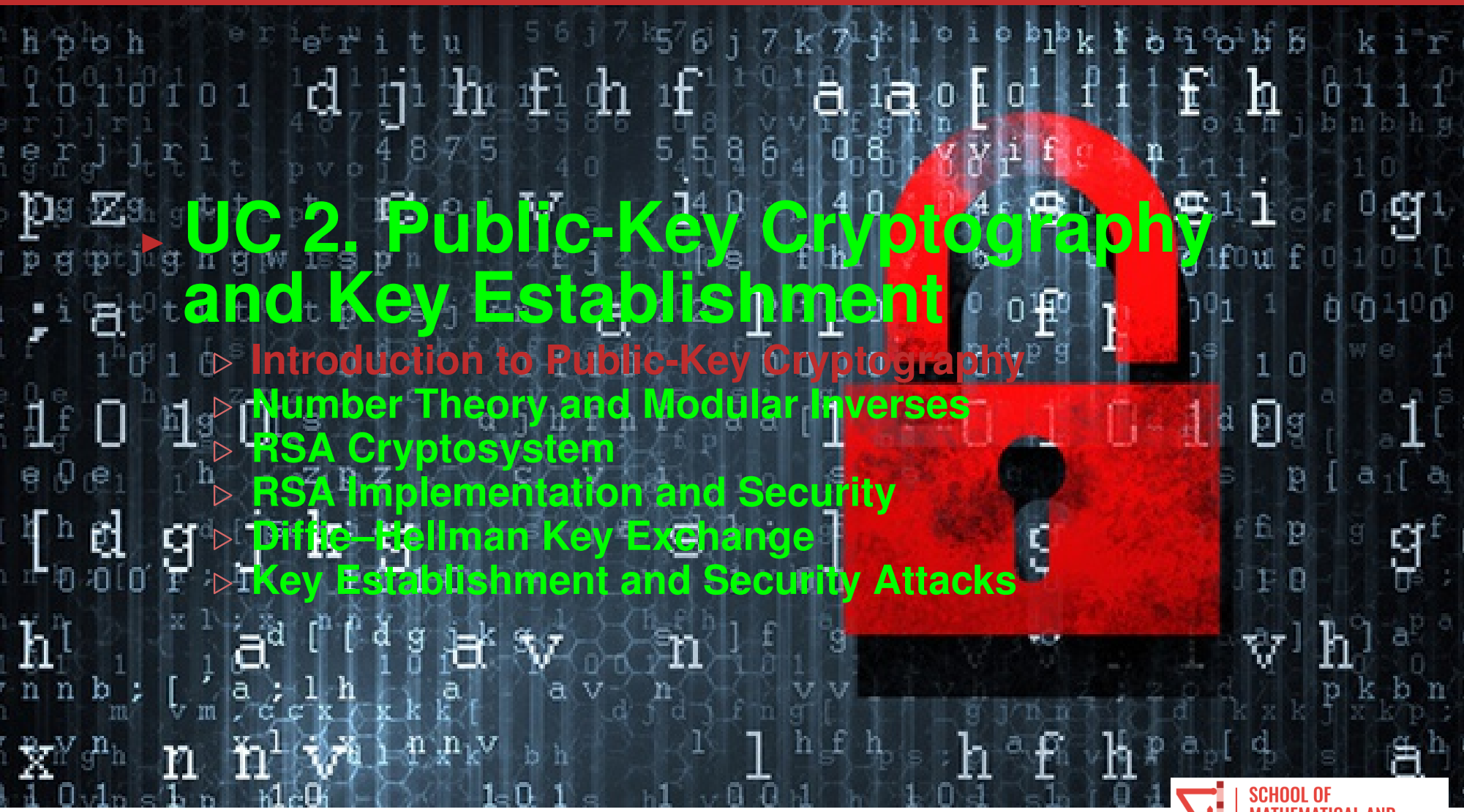
School of Mathematical and Computational Sciences
Yachay Tech University

September 16, 2026

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Number Theory

Public-Key Cryptography



UC 2. Public-Key Cryptography and Key Establishment

- Introduction to Public-Key Cryptography
- Number Theory and Modular Inverses
- RSA Cryptosystem
- RSA Implementation and Security
- Diffie–Hellman Key Exchange
- Key Establishment and Security Attacks

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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



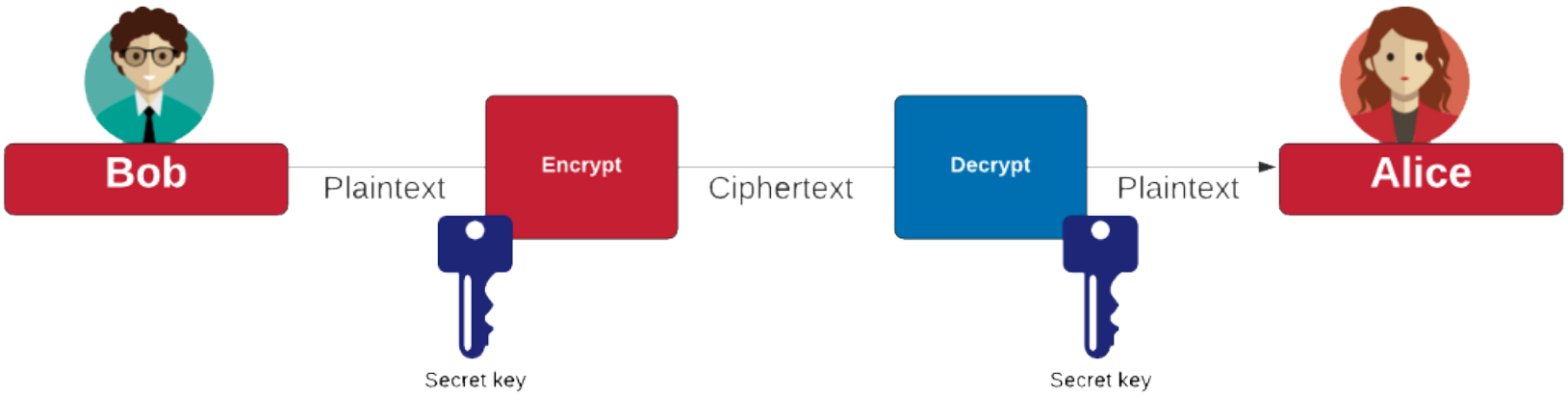
Key
interchange

Phil
Zimmerman



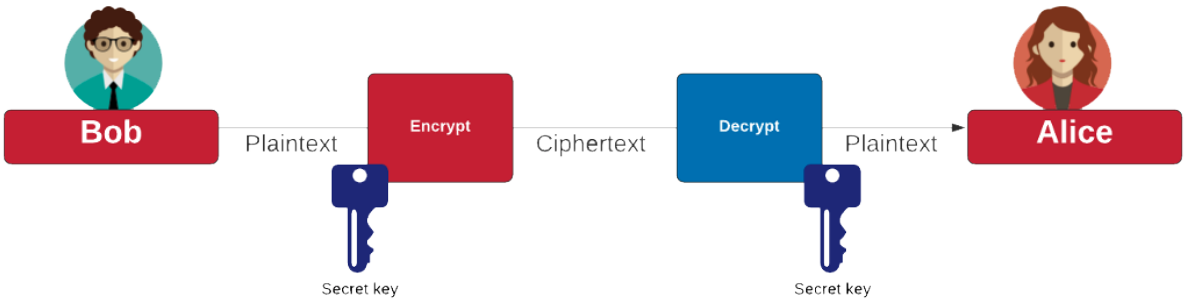
PGP
Encryption

Limitations of Symmetric Cryptography



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Limitations of Symmetric Cryptography



- **Key Distribution Problem:** Alice and Bob must securely share the secret key before communication.
- **Key Scalability:** for n users, the number of pairwise secret keys is

$$N_{\text{keys}} = \frac{n(n-1)}{2}.$$

- **No Non-Repudiation:** because both users know the same secret key, it is difficult to prove which party generated a message.

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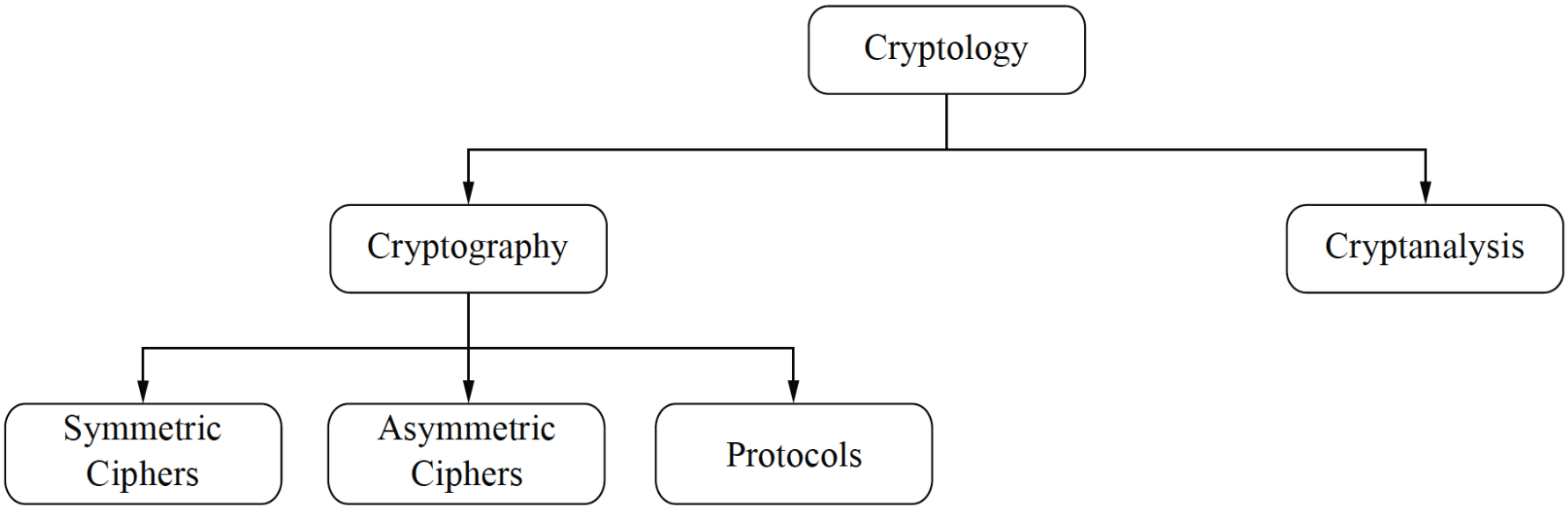
Limitations of Symmetric-Key Cryptography

- **Single Shared Key:** the same secret key is used for both encryption and decryption.
- **Key Distribution:** the sender and receiver must securely share the secret key before communication.
- **Key Compromise:** if the secret key is disclosed, the confidentiality of the communication is compromised.
- **Scalability Problem:** in a network with n users, each pair of users requires a different shared secret key.

$$N_{\text{keys}} = \frac{n(n-1)}{2}$$

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Introduction to Public-Key Cryptography

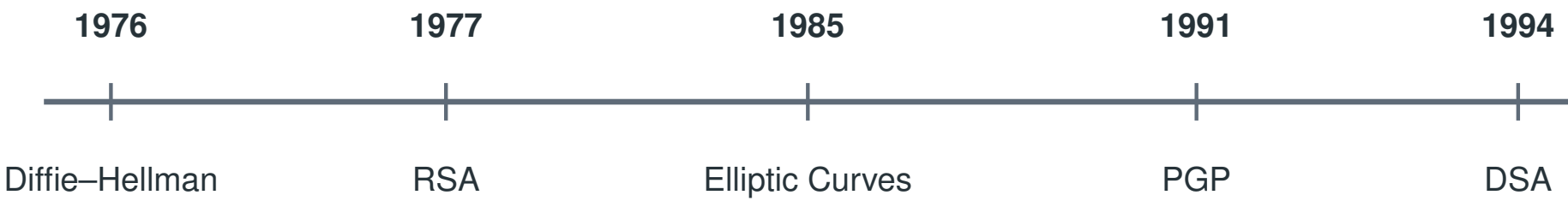


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Evolution of Public-Key Cryptography

1976: New concept
Public-key cryptography and secure key establishment

Digital signatures and public-key systems become practical security tools



Main families:
RSA → integer factorization
ECC → elliptic-curve discrete logarithm

Historical impact

Public-key cryptography changed secure communication by allowing **key establishment, encryption, and digital signatures** without requiring a previously shared secret key.

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Introduction to Public-Key Cryptography

Public-Key Cryptography

- Public-key cryptography is also known as **asymmetric cryptography**.
- It uses a pair of mathematically related keys:
  - **Public key:** can be shared with others.
  - **Private key:** must be kept secret.
- The concept was publicly introduced in 1976 by **Whitfield Diffie and Martin Hellman**, building on ideas also developed by **Ralph Merkle**.

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The Birth of Public-Key Cryptography

A New Direction in Cryptography

- In **1976**, Whitfield Diffie and Martin Hellman published the landmark paper:

  "New Directions in Cryptography"

- They introduced the concept of **public-key cryptography**.
- Their work proposed a new solution to the **key distribution problem**.

1976

A fundamental change in modern cryptography



644 IEEE TRANSACTIONS ON INFORMATION THEORY, VOL. IT-22, NO. 6, NOVEMBER 1976

New Directions in Cryptography

Invited Paper

WHITFIELD DIFFIE AND MARTIN E. HELLMAN, MEMBER, IEEE

Abstract—**Two kinds of contemporary developments in cryptography are examined. Widening applications of teleprocessing have given rise to a need for new types of cryptographic systems, which minimize the need for secure key distribution channels and supply the equivalent of a written signature. This paper suggests ways to solve these currently open problems. It also discusses how the theories of communication and computation are beginning to provide the tools to solve cryptographic problems of long standing.**

I. INTRODUCTION

WE STAND TODAY on the brink of a revolution in cryptography. The development of cheap digital hardware has freed it from the design limitations of mechanical computing and brought the cost of high grade cryptographic devices down to where they can be used in such commercial applications as remote cash dispensers and computer terminals. In turn, such applications create a need for new types of cryptographic systems which minimize the necessity of secure key distribution channels and supply the equivalent of a written signature. At the same time, theoretical developments in information theory and computer science show promise of providing provably secure cryptosystems, changing this ancient art into a science.

The development of computer controlled communication networks promises effortless and inexpensive contact between people or computers on opposite sides of the world, replacing most mail and many excursions with telecommunications. For many applications these contacts must be made secure against both eavesdropping and the injection of illegitimate messages. At present, however, the solution of security problems lags well behind other areas of communications technology. Contemporary cryptography is unable to meet the requirements, in that its use would impose such severe inconveniences on the system users, as to eliminate many of the benefits of teleprocessing.

Manuscript received June 3, 1976. This work was partially supported by the National Science Foundation under NSF Grant ENG 10173. Portions of this work were presented at the IEEE Information Theory Workshop, Lenox - MA, June 23–25, 1975 and the IEEE International Symposium on Information Theory in Ronneby, Sweden, June 21–24, 1976.

W. Diffie is with the Department of Electrical Engineering, Stanford University, Stanford, CA, and the Stanford Artificial Intelligence Laboratory, Stanford, CA 94305.

M. E. Hellman is with the Department of Electrical Engineering, Stanford University, Stanford, CA 94305.

The best known cryptographic problem is that of privacy: preventing the unauthorized extraction of information from communications over an insecure channel. In order to use cryptography to insure privacy, however, it is currently necessary for the communicating parties to share a key which is known to no one else. This is done by sending the key in advance over some secure channel such as private courier or registered mail. A private conversation between two people with no prior acquaintance is a common occurrence in business, however, and it is unrealistic to expect initial business contacts to be postponed long enough for keys to be transmitted by some physical means. The cost and delay imposed by this key distribution problem is a major barrier to the transfer of business communications to large teleprocessing networks.

Section III proposes two approaches to transmitting keying information over public (i.e., insecure) channels without compromising the security of the system. In a *public key cryptosystem* enciphering and deciphering are governed by distinct keys, E and D, such that computing D from E is computationally infeasible (e.g., requiring 10^{100} instructions). The enciphering key E can thus be publicly disclosed without compromising the deciphering key D. Each user of the network can, therefore, place his enciphering key in a public directory. This enables any user of the system to send a message to any other user enciphered in such a way that only the intended receiver is able to decipher it. As such, a public key cryptosystem is a multiple access cipher. A private conversation can therefore be held between any two individuals regardless of whether they have ever communicated before. Each one sends messages to the other enciphered in the receiver's public enciphering key and deciphers the messages he receives using his own secret deciphering key.

We propose some techniques for developing public key cryptosystems, but the problem is still largely open.

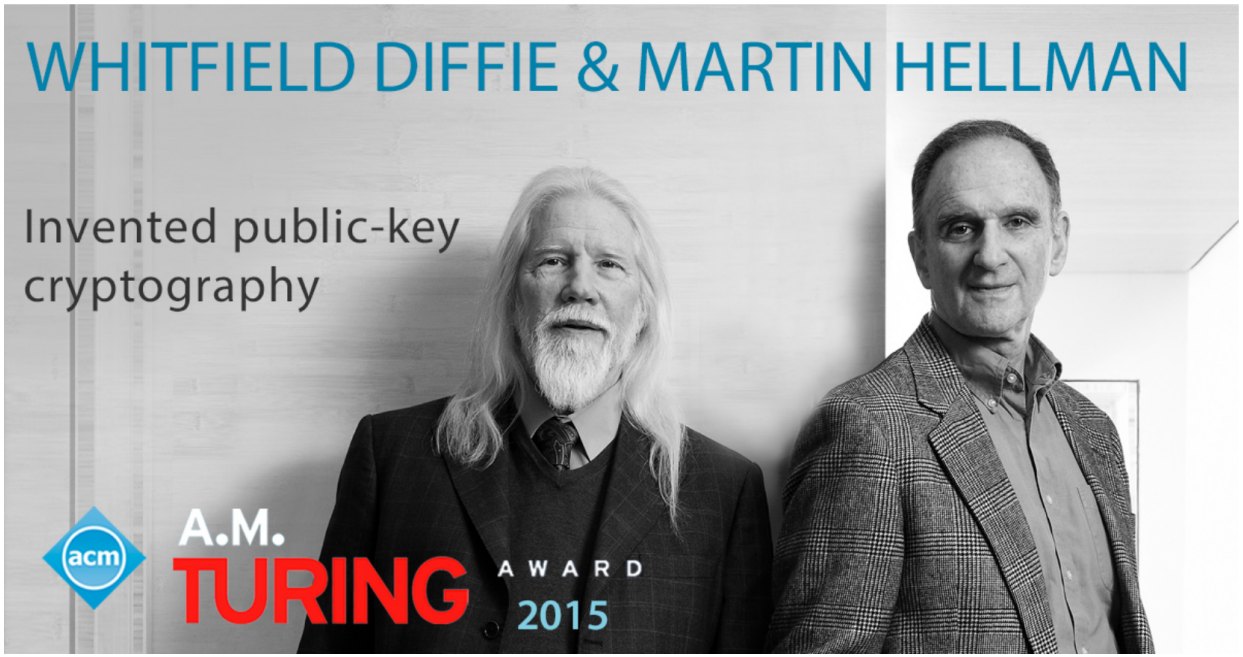
Public key distribution systems offer a different approach to eliminating the need for a secure key distribution channel. In such a system, two users who wish to exchange a key communicate back and forth until they arrive at a key in common. A third party eavesdropping on this exchange must find it computationally infeasible to compute the key from the information overheard. A possible solution to the public key distribution problem is given in Section III, and Merkle [1] has a partial solution of a different form.

A second problem, amenable to cryptographic solution, which stands in the way of replacing contemporary busi-

W. Diffie and M. E. Hellman,IEEE Transactions on Information Theory, Vol. 22, No. 6, 1976.

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2015 ACM A.M. Turing Award



- **Whitfield Diffie** and **Martin Hellman** received the **2015 ACM A.M. Turing Award**.
- The award recognized their fundamental contributions to **modern cryptography**.
- Their work introduced the principles of **public-key cryptography** and **digital signatures**.
- These ideas changed how secure communication can be established over an **insecure network**.

1976 $\longrightarrow$ **2015** From a new cryptographic concept to a major recognition in computer science

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Main Security Mechanisms of Public-Key Cryptography

1. Key Establishment

Establish a **shared secret key** over an insecure network.

Examples: Diffie–Hellman, RSA key transport.

2. Digital Signatures

Provide **authentication**, **integrity**, and support **nonrepudiation**.

Examples: RSA, DSA, ECDSA.

3. Identification

Verify the identity of an entity using cryptographic **challenge–response protocols**.

Applications: Smart cards, secure devices, authentication systems.

4. Encryption

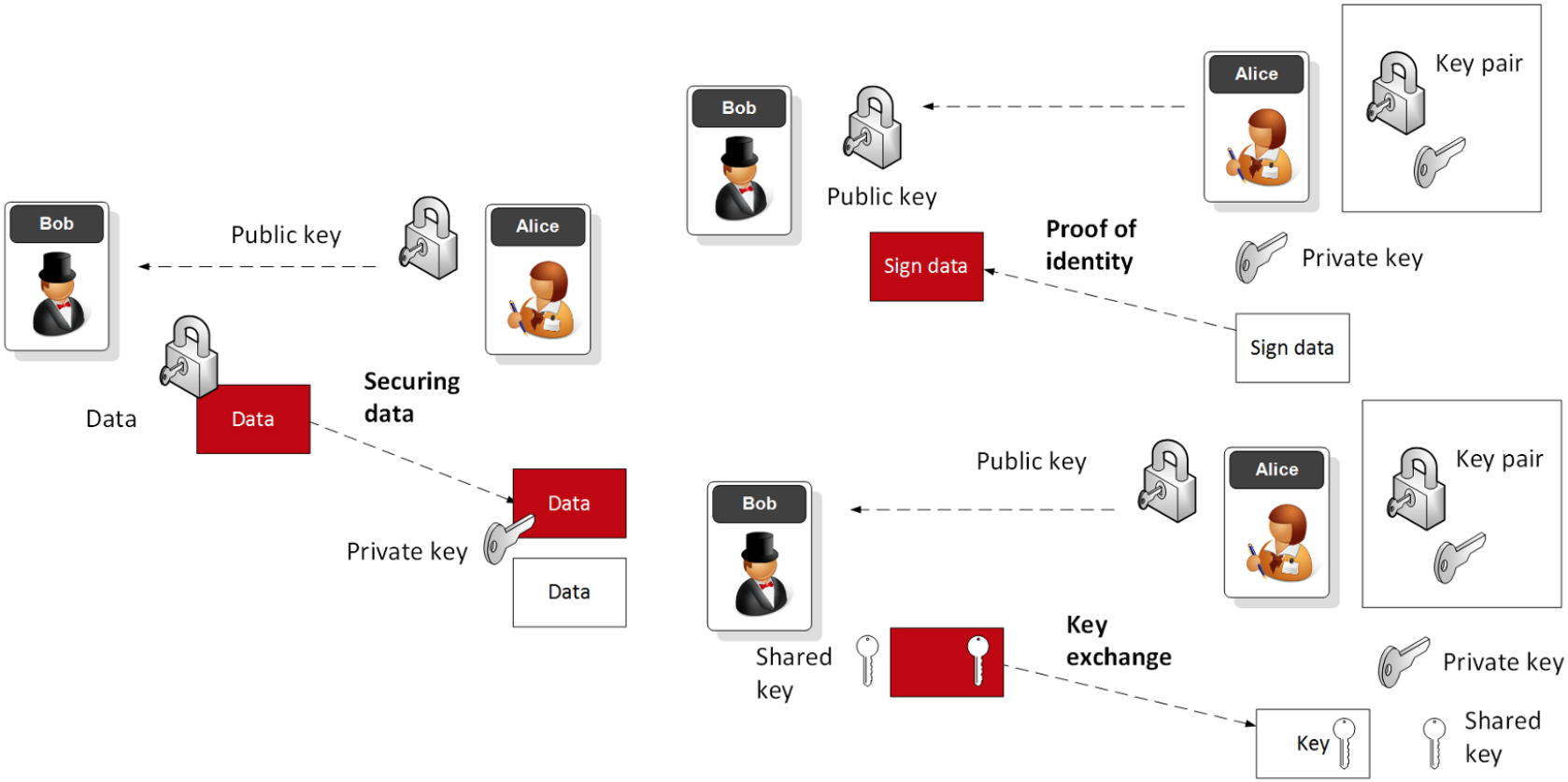
Protect the **confidentiality** of information using the recipient's public key.

Examples: RSA, ElGamal.

Public-key cryptography enables secure communication without a previously shared secret key.

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Main Security Mechanisms of Public-Key Algorithms



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Public-Key Algorithm Families

Public-Key Algorithm Families of Practical Relevance

Integer-Factorization Schemes Several public-key schemes are based on the fact that it is difficult to factor large integers. The most prominent representative of this algorithm family is RSA.

Discrete Logarithm Schemes There are several algorithms which are based on what is known as the discrete logarithm problem in finite fields. The most prominent examples include the Diffie–Hellman key exchange, Elgamal encryption or the Digital Signature Algorithm (DSA).

Elliptic Curve (EC) Schemes A generalization of the discrete logarithm algorithm are elliptic curve public-key schemes. The most popular examples include Elliptic Curve Diffie–Hellman key exchange (ECDH) and the Elliptic Curve Digital Signature Algorithm (ECDSA).

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Public-Key Algorithm Families

Integer Factorization

Hard Problem

Factor a large integer

$$n = pq$$

Main example
RSA

Applications
Encryption
Digital signatures

Discrete Logarithm

Hard Problem

Given

$$g^x \equiv y \pmod{p}$$

find x

Main examples
Diffie–Hellman
ElGamal
DSA

Applications
Key establishment
Encryption
Digital signatures

Elliptic Curves

Hard Problem

Given

$$Q = kP$$

find k

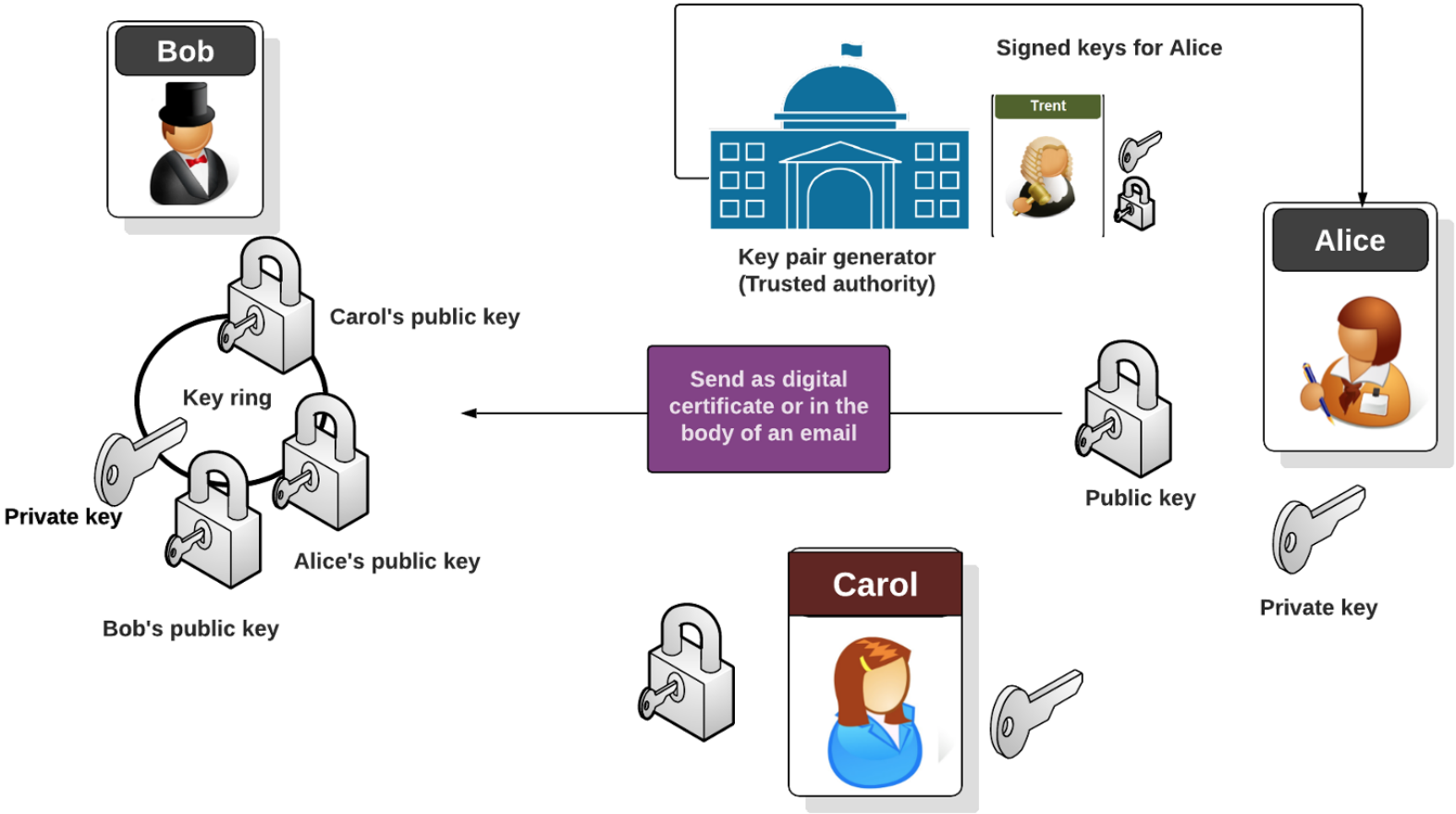
Main examples
ECDH
ECDSA

Applications
Key establishment
Digital signatures

The security of each family relies on a computationally hard mathematical problem.

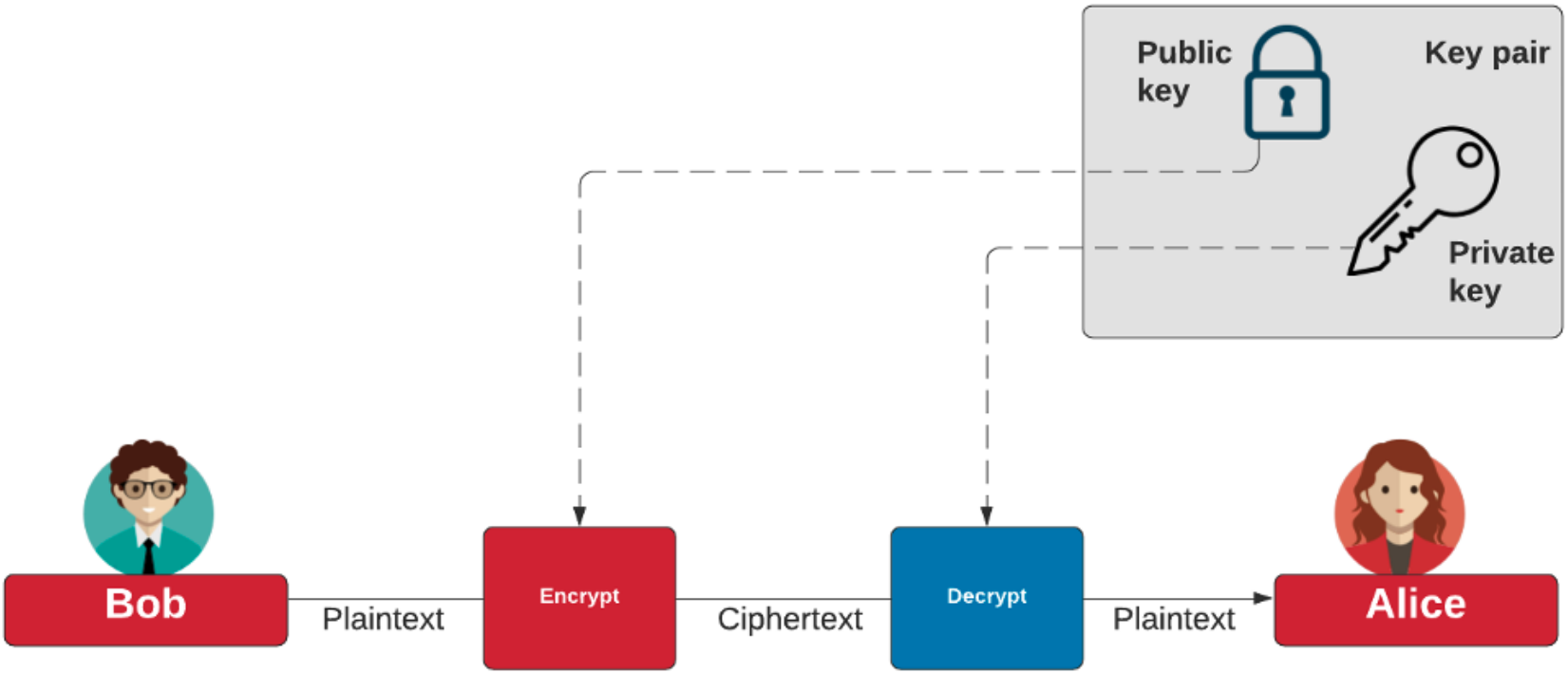
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The Remaining Problem: Authenticity of Public Keys



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Secure Use of Asymmetric Encryption

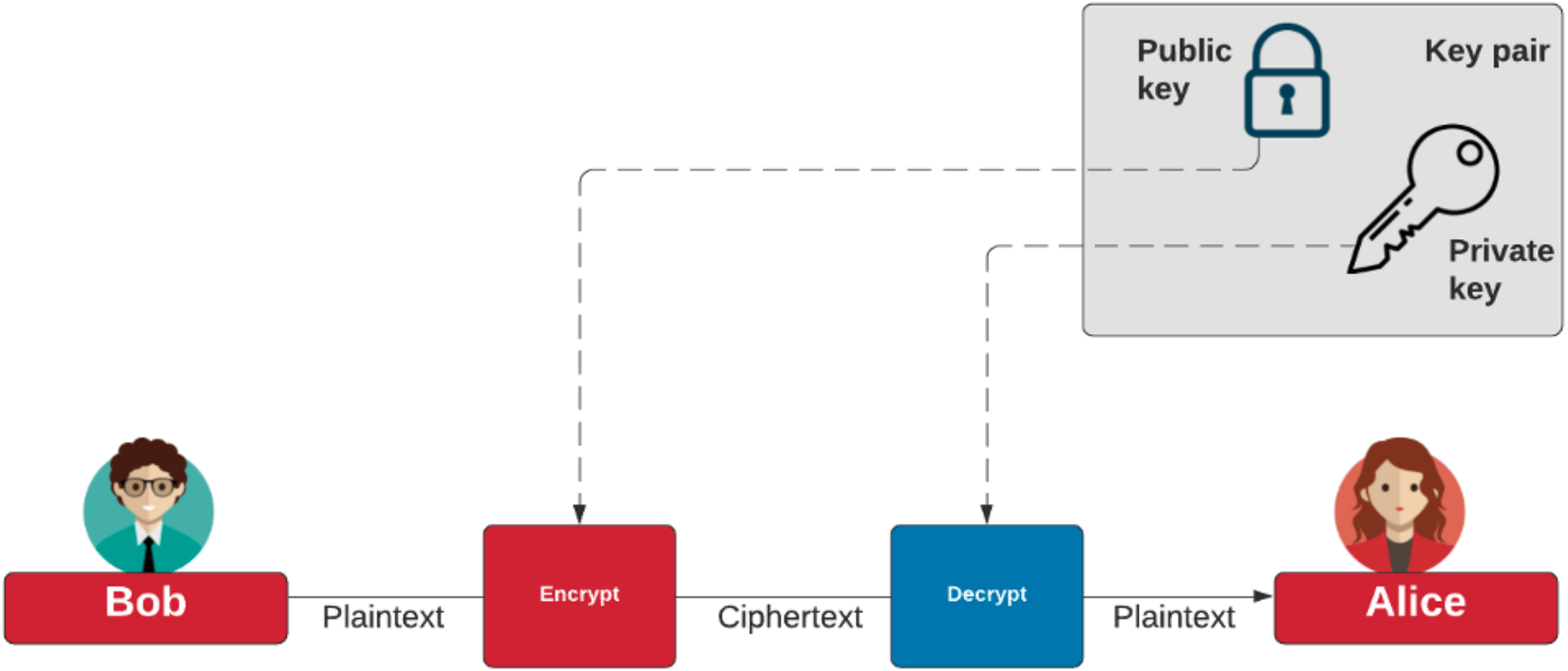


1 Key Pair Generation

- Alice generates a pair of **mathematically related keys**: a public key and a private key.
- The **public key** can be distributed openly to users who want to send encrypted data to Alice.
- The **private key** must remain secret and under Alice's control.

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Secure Use of Asymmetric Encryption



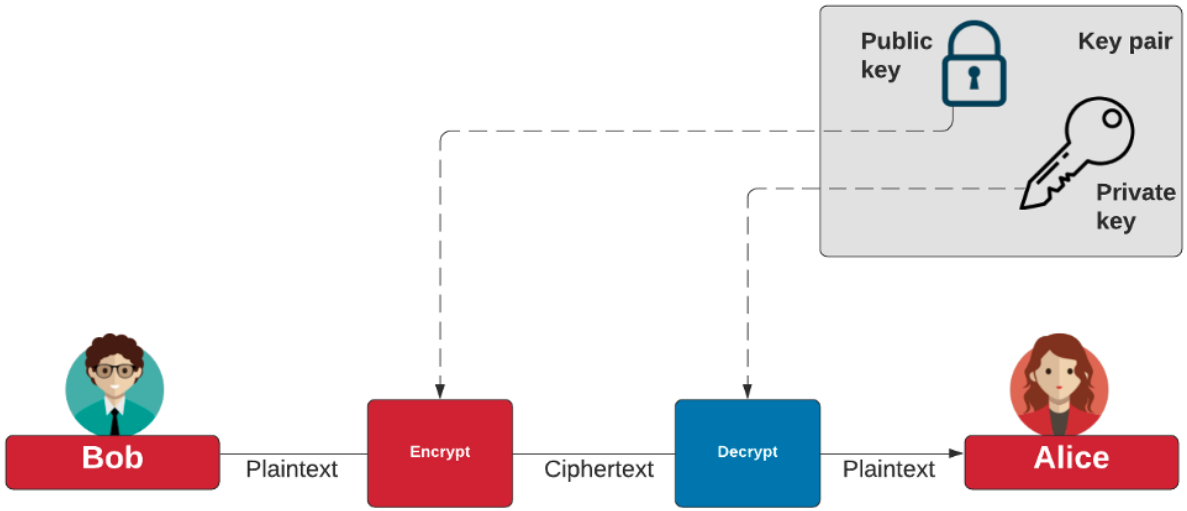
2 Encryption

- Bob obtains **Alice's public key**.
- Bob uses Alice's public key to encrypt the plaintext message.
- The encryption process transforms the plaintext into **ciphertext**.
- The ciphertext can be decrypted only with the corresponding **private key**, which is kept by Alice.

$$C = E(K_{PU_A}, M)$$

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Secure Use of Asymmetric Encryption



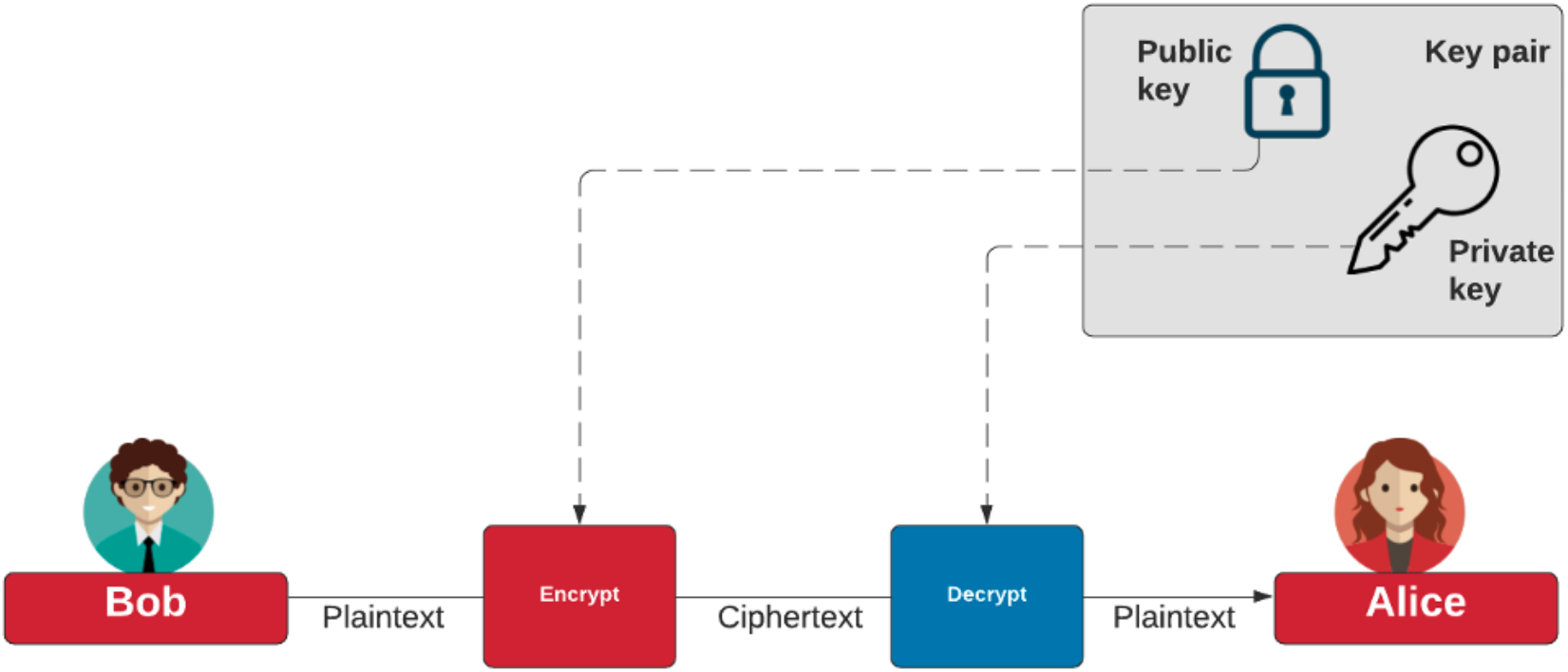
3 Decryption

- Alice receives the **ciphertext** sent by Bob.
- Alice uses her corresponding **private key** to decrypt the ciphertext.
- The decryption process recovers the original **plaintext message**.
- Only the holder of the corresponding private key should be able to recover the plaintext.

$$M = D(K_{PR_A}, C)$$

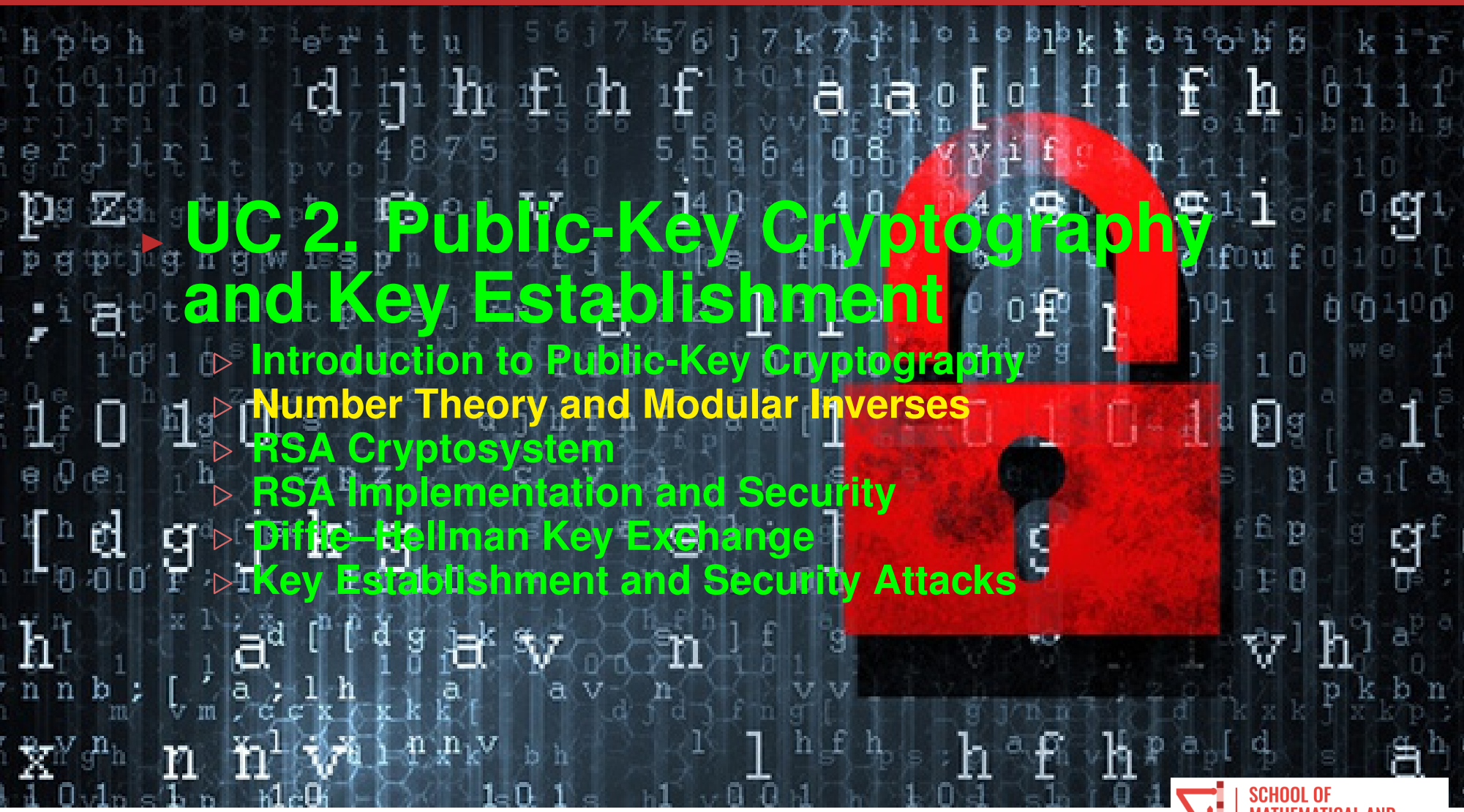
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Public Key Encryption



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Public-Key Cryptography



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Bruce Schneier



Twofish,
Blowfish

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AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



Key
interchange

Phil
Zimmerman



PGP
Encryption

Public-Key Cryptography

Number Theory for Public-Key Algorithms

- Euclidean Algorithm
- Euler's phi function
- Fermat's Little Theorem
- Euler's theorem

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Euclidean Algorithm

Greatest Common Divisor

- The **greatest common divisor** of two positive integers r_0 and r_1 is the largest positive integer that divides both numbers.

$$\gcd(r_0, r_1)$$

Example: Let $r_0 = 84$ and $r_1 = 30$.

$$84 = 2^2 \cdot 3 \cdot 7 \qquad 30 = 2 \cdot 3 \cdot 5$$

The common prime factors are 2 and 3:

$$\boxed{\gcd(84, 30) = 2 \cdot 3 = 6}$$

Factoring is simple for small numbers, but inefficient for large integers.

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Euclidean Algorithm

- For two positive integers $r_0 > r_1$:

$$\boxed{\gcd(r_0, r_1) = \gcd(r_0 - r_1, r_1)}$$

- Subtracting the smaller number does **not change the greatest common divisor**.

Example: $r_0 = 84, \qquad r_1 = 30$

$$84 - 30 = 54$$

$$\gcd(84, 30) = \gcd(54, 30) = \boxed{6}$$

The same operation can be repeated until the numbers become smaller.

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Euclidean Algorithm: From Subtraction to Modulo (cont.)

- Repeated subtraction does not change the greatest common divisor:

$$\begin{aligned}\gcd(r_0, r_1) &= \gcd(r_0 - r_1, r_1)\\ &= \gcd(r_0 - 2r_1, r_1)\\ &= \dots\\ &= \gcd(r_0 - mr_1, r_1)\end{aligned}$$

and therefore

$$r_0 - mr_1 = r_0 \bmod r_1.$$

$$\boxed{\gcd(r_0, r_1) = \gcd(r_1, r_0 \bmod r_1)}$$

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Euclidean Algorithm: From Subtraction to Modulo (cont.)

Recursive Rule

Apply the same operation repeatedly:

$$\gcd(r_0, r_1) \to \gcd(r_1, r_2) \to \gcd(r_2, r_3) \to \cdots$$

where

$$r_{i+1} = r_{i-1} \bmod r_i.$$

When the remainder becomes zero, the previous remainder is the gcd.

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Number Theory for Public-Key Algorithms

Euclidean Algorithm

Let $r_0 = 973$ and $r_1 = 301$. The gcd is computed as:

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Euclidean Algorithm (cont.)

Let $r_0 = 973$ and $r_1 = 301$. The gcd is computed as:

Step	Dividend	Divisor	Remainder
1	973	301	70
2	301	70	21
3	70	21	7
4	21	7	0

$$\begin{aligned} 973 &= 3(301) + 70 \\ 301 &= 4(70) + 21 \\ 70 &= 3(21) + 7 \\ 21 &= 3(7) + 0 \end{aligned}$$

The last nonzero remainder is the gcd.

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Number Theory for Public-Key Algorithms

Euclidean Algorithm

Let $r_0 = 973$ and $r_1 = 301$. The gcd is computed as:

$$\begin{aligned} gcd(973, 301) &= \gcd(973 \bmod 301, 301) \\ &= \gcd(70, 301) \\ &= \gcd(301 \bmod 70, 70) \\ &= \gcd(21, 70) \\ &= \gcd(70 \bmod 21, 21) \\ &= \gcd(21, 7) \\ &= \gcd(7, 21 \bmod 7) \\ \gcd(r_0, r_1) &= \gcd(7, 0) = 7 \end{aligned}$$

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Number Theory for Public-Key Algorithms

Euclidean Algorithm
Input: positive integers r_0 and r_1 with $r_0 > r_1$
Output: $\gcd(r_0, r_1)$
Initialization: $i = 1$
Algorithm:

1 DO
1.1 $i = i + 1$
1.2 $r_i = r_{i-2} \bmod r_{i-1}$
WHILE $r_i \neq 0$
2 RETURN
$\gcd(r_0, r_1) = r_{i-1}$

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Exercise 1: Euclidean Algorithm

Compute:

$$\boxed{\gcd(899, 493)}$$

Apply the Euclidean Algorithm:

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Exercise 1: Euclidean Algorithm

Compute:

$$\boxed{\gcd(899, 493)}$$

Apply the Euclidean Algorithm:

$$\begin{aligned} 899 &= 1(493) + 406 \\ 493 &= 1(406) + 87 \\ 406 &= 4(87) + 58 \\ 87 &= 1(58) + 29 \\ 58 &= 2(29) + 0 \end{aligned}$$

The last nonzero remainder is 29. $\rightarrow$ gcd(899,493)=29

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Exercise 2: Euclidean Algorithm

Compute:

$$\boxed{\gcd(3120, 17)}$$

Apply the Euclidean Algorithm:

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Exercise 2: Euclidean Algorithm

Compute:

$$\boxed{\gcd(3120, 17)}$$

Apply the Euclidean Algorithm:

$$\begin{aligned} 3120 &= 183(17) + 9 \\ 17 &= 1(9) + 8 \\ 9 &= 1(8) + 1 \\ 8 &= 8(1) + 0 \end{aligned}$$

$$\boxed{\gcd(3120, 17) = 1}$$

Therefore, 3120 and 17 are **coprime**.

$$\boxed{17^{-1} \pmod{3120} \text{ exists}}$$

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Extended Euclidean Algorithm

- The **Extended Euclidean Algorithm (EEA)** computes the **gcd** and two integer coefficients s and t such that:

$$\boxed{\gcd(r_0, r_1) = s\, r_0 + t\, r_1}$$

This equation is called **Bézout's identity**.

- The coefficients $s, t \in \mathbb{Z}$ are called **Bézout coefficients**.
- If $gcd(r_0, r_1) = 1$, the EEA can be used to compute a **modular inverse**.

How do we compute the coefficients s and t?

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Number Theory for Public-Key Algorithms

$$\gcd(973, 301) = s \cdot 973 + t \cdot 301$$

(973, 301)	973=3 · 301 + 70	70=973-3·301
(301, 70)	301=4 · 70 + 21	21= 301-4·70
(70, 21)	70= 3 · 21 + 7	**7 = 70 - 3 · 21**
(21, 7)	21 = 3 · 7 + 0	7 = (973-3·301) - 3 (301-4·70)
(7, 0)		7 = (973-3·301) - 3 (301-4·(973-3·301))
		7= 973 - 3·301 - 3· 301 +12(973 - 3·301)
		7=973 - 3·301 - 3· 301 +12· 973 - 36·301
		7=13 · 973 - 42 · 301

$$\gcd(r_0, r_1) = 13 \cdot r_0 - 42 \cdot r_1$$

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Extended Euclidean Algorithm: Back Substitution (cont.)

From the Euclidean Algorithm:

$$\begin{aligned} 973 &= 3(301) + 70 \\ 301 &= 4(70) + 21 \\ 70 &= 3(21) + 7 \end{aligned}$$

Now solve each equation for the remainder and substitute backwards:

$$\begin{aligned} 7 &= 70 - 3(21) \\ &= 70 - 3(301 - 4(70)) \\ &= 13(70) - 3(301) \\ &= 13(973 - 3(301)) - 3(301) \\ &= \boxed{13(973) - 42(301)} \end{aligned}$$

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Extended Euclidean Algorithm: Back Substitution (cont.)

Therefore,

$$\boxed{\gcd(973, 301) = 7 = 13(973) - 42(301)}$$

Bézout coefficients: $s = 13, \qquad t = -42$

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Extended Euclidean Algorithm and Modular Inverse

- A main application of the **Extended Euclidean Algorithm (EEA)** is to compute a modular inverse.
- A modular inverse exists only if $gcd(r_0, r_1) = 1$

In this case, Bézout's identity becomes:

$$sr_0 + tr_1 = 1$$

Taking both sides modulo r_0:

$$\begin{aligned} sr_0 + tr_1 &\equiv 1 \pmod{r_0} \\ tr_1 &\equiv 1 \pmod{r_0} \end{aligned}$$

$$\therefore \boxed{t \equiv r_1^{-1} \pmod{r_0}}$$

The Bézout coefficient t gives the inverse of r_1 modulo r_0.

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Extended Euclidean Algorithm: Modular Inverse (cont.)

- To compute the modular inverse

$$a^{-1} \pmod{m},$$

first check that:

$$\boxed{\gcd(a, m) = 1}$$

- Then apply the **Extended Euclidean Algorithm** to find integers s and t such that:

$$sm + ta = 1$$

- The coefficient of a gives the modular inverse:

$$\boxed{a^{-1} \equiv t \pmod{m}}$$

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Extended Euclidean Algorithm: Modular Inverse (cont.)

Example

Compute:

$$12^{-1} \pmod{67}$$

Extended Euclidean Algorithm

Compute the modular inverse:

$$37^{-1} \pmod{101}$$

Extended Euclidean Algorithm

Compute the modular inverse:

$$17^{-1} \pmod{3120}$$

Euler's Phi Function

Euler's Phi Function

- Consider the set of integers modulo m: $\mathbb{Z}_m = \{0, 1, 2, \ldots, m-1\}$
- We want to count the elements that are **relatively prime to** m.

When is an integer a relatively prime to m?

$$\boxed{\gcd(a, m) = 1}$$

- The number of integers in $\mathbb{Z}_m$ that satisfy this condition is called **Euler's Phi Function**.

$$\boxed{\phi(m) = \#\{\, a \in \mathbb{Z}_m : \gcd(a, m) = 1 \,\}}$$

$\phi(m)$ counts how many elements have a multiplicative inverse modulo m.

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Euler's Phi Function: Example (cont.)

Let $m = 6$. Then: $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

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Euler's Phi Function: Example (cont.)

Let $m = 6$. Then: $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

a	$\gcd(a, 6)$	**Relatively prime to 6?**
0	6	No
1	1	**Yes**
2	2	No
3	3	No
4	2	No
5	1	**Yes**

The elements relatively prime to 6 are $\boxed{\{1, 5\}}$

Therefore $\boxed{\phi(6) = 2}$

There are 2 invertible elements in $\mathbb{Z}_6$.

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Euler's Phi Function: Prime Modulus

Let $m = 5$. Then $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

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Euler's Phi Function: Prime Modulus

Let $m = 5$. Then $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

a	$\gcd(a, 5)$	**Relatively prime to 5?**
0	5	No
1	1	**Yes**
2	1	**Yes**
3	1	**Yes**
4	1	**Yes**

The elements relatively prime to 5 are $\boxed{\{1, 2, 3, 4\}}$

Therefore $\boxed{\phi(5) = 4}$

If p is prime, then $\phi(p) = p - 1$.

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Number Theory for Public-Key Algorithms (cont.)

Theorem

Let m have the following canonical factorization

$$m = p_1^{e_1} \cdot p_2^{e_2} \cdots p_n^{e_n}$$

where the p_i are distinct prime numbers and e_i are positive integers, then

$$\phi(m) = \prod_{i=1}^{n} \left(p_i^{e_i} - p_i^{e_i - 1} \right)$$

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Euler's Phi Function: General Formula (cont.)

Step 1: Prime Factorization

Let m have the prime factorization:

$$\boxed{m = p_1^{e_1} p_2^{e_2} \cdots p_n^{e_n}}$$

where:

- $p_1, p_2, \ldots, p_n$ are **distinct prime numbers**,
- $e_1, e_2, \ldots, e_n$ are **positive integers**.

The goal is to count the integers in $\mathbb{Z}_m$ that are **relatively prime to** m.

$$\phi(m) = \#\{\, a \in \mathbb{Z}_m : \gcd(a, m) = 1 \,\}$$

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Euler's Phi Function: General Formula (cont.)

Step 2: Consider a Prime Power

First, consider:

$$m = p^e$$

There are p^e elements in $\mathbb{Z}_{p^e}$.

The elements that are **not relatively prime** to p^e are the multiples of p.
The number of multiples of p is:

$$\frac{p^e}{p} = p^{e-1}$$

Therefore:

$$\phi(p^e) = \underbrace{p^e}_{\text{all elements}} - \underbrace{p^{e-1}}_{\text{multiples of } p}$$

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Euler's Phi Function: General Formula (cont.)

$$\boxed{\phi(p^e) = p^e - p^{e-1}}$$

Factoring p^e:

$$\boxed{\phi(p^e) = p^e \left(1 - \frac{1}{p}\right)}$$

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Euler's Phi Function: General Formula (cont.)

Step 3: Combine the Prime Powers

The prime powers

$$p_1^{e_1},\ p_2^{e_2},\ \ldots,\ p_n^{e_n}$$

are pairwise relatively prime.
Therefore, Euler's Phi Function is **multiplicative**:

$$\phi(ab) = \phi(a)\phi(b) \qquad \text{if } \gcd(a,b) = 1$$

Thus:

$$\phi(m) = \phi(p_1^{e_1})\phi(p_2^{e_2})\cdots\phi(p_n^{e_n})$$

Substituting

$$\phi(p_i^{e_i}) = p_i^{e_i} - p_i^{e_i-1},$$

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Euler's Phi Function: General Formula (cont.)

we obtain:

$$\boxed{\phi(m) = \prod_{i=1}^{n} \left(p_i^{e_i} - p_i^{e_i - 1} \right)}$$

Step 4: Equivalent Formula

Since:

$$p_i^{e_i} - p_i^{e_i - 1} = p_i^{e_i} \left(1 - \frac{1}{p_i} \right),$$

then:

$$\phi(m) = \prod_{i=1}^{n} p_i^{e_i} \left(1 - \frac{1}{p_i} \right)$$

But:

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Euler's Phi Function: General Formula (cont.)

$$\prod_{i=1}^{n} p_i^{e_i} = m$$

Therefore:

$$\boxed{\phi(m) = m \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right)}$$

Euler's Phi Function counts the integers that are relatively prime to m.

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Euler's Phi Function: Using Prime Powers

Let $m = 240$. Prime factorization $\boxed{240 = 2^4 \cdot 3 \cdot 5}$

$$\boxed{\phi(m) = m \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right)}$$

Using $\phi(m) = \prod_{i=1}^{n} \left(p_i^{e_i} - p_i^{e_i - 1}\right)$,
we obtain:

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Euler's Phi Function: Using Prime Powers

Let $m = 240$. Prime factorization $\boxed{240 = 2^4 \cdot 3 \cdot 5}$

Thus, $p_1 = 2, \quad p_2 = 3, \quad p_3 = 5$

and $e_1 = 4, \quad e_2 = 1, \quad e_3 = 1$

Using $\phi(m) = \prod_{i=1}^{n} \left(p_i^{e_i} - p_i^{e_i - 1} \right)$,

we obtain:

$$\begin{aligned} \phi(240) &= (2^4 - 2^3)(3^1 - 3^0)(5^1 - 5^0) \\ &= (16 - 8)(3 - 1)(5 - 1) \\ &= 8 \cdot 2 \cdot 4 \\ &= \boxed{64} \end{aligned}$$

There are 64 integers in $\mathbb{Z}_{240}$ relatively prime to 240.

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Euler's Phi Function: Equivalent Formula (cont.)

From the prime factorization:

$$240 = 2^4 \cdot 3 \cdot 5$$

the distinct prime factors are:

$$\boxed{2,\ 3,\ 5}$$

Using the equivalent formula:

$$\boxed{\phi(m) = m \prod_{p|m} \left(1 - \frac{1}{p}\right)}$$

we obtain:

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Euler's Phi Function: Equivalent Formula (cont.)

$$\begin{aligned}\phi(240) &= 240\left(1-\frac{1}{2}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{5}\right)\\ &= 240\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(\frac{4}{5}\right)\\ &= \boxed{64}\end{aligned}$$

Both formulas give the same result: $\phi(240) = 64$.

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Fermat's Little

Fermat's Little Theorem (cont.)

Let p be a **prime number** and let a be an integer such that

$$\gcd(a, p) = 1.$$

Then:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}}$$

- Since p is prime, the condition is equivalent to:

$$p \nmid a$$

- Fermat's Little Theorem connects **prime numbers** with **modular arithmetic**.

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Fermat's Little Theorem (cont.)

Computing a Modular Inverse

From

$$a^{p-1} \equiv 1 \pmod{p},$$

we obtain:

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

Therefore:

$$\boxed{a^{-1} \equiv a^{p-2} \pmod{p}}$$

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Fermat's Little Theorem: Example (cont.)

Compute:

$$\boxed{2^{-1} \pmod 7}$$

Since 7 is prime and

$$\gcd(2, 7) = 1,$$

Fermat's Little Theorem gives:

$$a^{-1} \equiv a^{p-2} \pmod p.$$

Therefore:

$$\begin{aligned} 2^{-1} &\equiv 2^{7-2} \pmod 7 \\ &\equiv 2^5 \pmod 7 \\ &\equiv 32 \pmod 7 \\ &\equiv \boxed{4} \pmod 7 \end{aligned}$$

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Fermat's Little Theorem: Example (cont.)

Verification

$$2 \cdot 4 = 8 \equiv 1 \pmod{7}$$

Therefore, $2^{-1} \equiv 4 \pmod{7}$**.**

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Euler's Theorem

Euler's Theorem (cont.)

Euler's Theorem generalizes Fermat's Little Theorem to a modulus m that does **not need to be prime**.

Let a and m be positive integers such that:

$$\boxed{\gcd(a, m) = 1}$$

Then:

$$\boxed{a^{\phi(m)} \equiv 1 \pmod{m}}$$

where $\phi(m)$ is Euler's Phi Function.

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Euler's Theorem (cont.)

Fermat vs. Euler

$$\text{Fermat} \quad : \quad a^{p-1} \equiv 1 \pmod{p}$$

$$\text{Euler} \quad : \quad a^{\phi(m)} \equiv 1 \pmod{m}$$

Euler's Theorem works for any modulus m if $\gcd(a, m) = 1$.

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Euler's Theorem: Example (cont.)

Let:

$$a = 5, \qquad m = 12.$$

Step 1: Check the condition

$$\gcd(5, 12) = 1$$

Therefore, Euler's Theorem can be applied.

Step 2: Compute $\phi(12)$
Since

$$12 = 2^2 \cdot 3,$$

then:

$$\begin{aligned} \phi(12) &= (2^2 - 2^1)(3^1 - 3^0) \\ &= (4 - 2)(3 - 1) \\ &= \boxed{4} \end{aligned}$$

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Euler's Theorem: Verification

We know that:

$$\phi(12) = 4$$

Euler's Theorem states:

$$5^{\phi(12)} \equiv 1 \pmod{12}.$$

Therefore:

$$5^4 = 625$$
$$625 \bmod 12 = 1$$

Thus:

$$\boxed{5^4 \equiv 1 \pmod{12}}$$

Euler's Theorem is verified.

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Cryptography

Public-Key Cryptography

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